

AttentionOS

The AI-Powered WhatsApp Message Notification Router

Product Requirements Document — Version 1.0

Classification: Confidential — Internal Product Proposal

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Date: August 1, 2026

Note on sourcing: inline citation-style markers from the original draft (referencing unresolvable shorthand tags) have been removed rather than reproduced as verified sources. One statistic — Meta's account-takedown figure — is corrected below to match public reporting. Market-size and cognitive-load figures elsewhere in this document (e.g. attention-economy valuation, notification-market CAGR, the 3–5 daily notification threshold) are treated as illustrative product assumptions, not verified research, pending a source. A hackathon Implementation Status appendix is included at the end.

1. Executive Summary

AttentionOS is framed not as a notification classifier but as a Digital Attention Operating System that models human attention as a finite resource. Every WhatsApp message is evaluated for its attention cost relative to the user's cognitive state, relationship context, and a daily attention budget, rather than for content alone.

The core mechanism is an Attention Budget: a small daily ceiling on high-priority notifications, enforced as a hard constraint so upstream reasoning must prioritize ruthlessly rather than notify by default. Messages are routed through a multimodal pipeline (text, image, voice) that combines relationship context, trust signals, and historical memory into an explainable notify / digest / mute decision.

The product targets three problems at once: notification fatigue, scam exposure, and lack of personalization — treating attention explicitly as a budget to be spent, not an inbox to be filled.

2. Product Vision

To become the layer between humans and digital communication that understands attention well enough to route every message to the right moment, in the right form, with a clear explanation for why.

3. Mission Statement

Protect human attention from digital chaos by reasoning over context, relationships, and cognitive load to deliver only what matters, when it matters.

4. Product Philosophy

Principle	Description
Attention is Finite	Notification volume should be treated as a limited daily budget, not an unlimited stream.
Context is King	A message's importance depends on who sent it, when, and what the user is doing.
Explainability Builds Trust	Every decision states why: notify, digest, or mute.
Relationships Matter	Family, work, and strangers warrant different treatment.
Safety First	Scams and fraud must be caught before they reach the user.
Adaptive Learning	The system should learn from every dismiss, act, and silence.

5. Problem Analysis

The Attention Crisis: users facing high notification volume across fragmented tools report rising cognitive overload and decision fatigue, which can cause important messages — urgent family updates, payment reminders — to be missed amid noise.

The Scam Epidemic: Meta reported removing 6.8 million WhatsApp accounts tied to criminal scam centers in the first half of 2025, and separately disrupted close to 8 million scam-linked accounts across Facebook and Instagram — underscoring that fraud on messaging platforms operates at meaningful scale. Common vectors include job scams, KYC fraud, UPI phishing, and fake investment schemes, and keyword-only filters routinely miss context-dependent or multimodal fraud (images, voice notes).

The Personalization Gap: current notification systems largely treat all users identically, without modeling individual attention limits or relationship context — no system today asks why a message matters to this specific user at this specific moment.

6. Existing Industry Problems

Problem	Current Approach	Why It Fails
Notification Overload	Priority inboxes, mute buttons	Reactive, not predictive; requires manual configuration
Scam Detection	Keyword filters, user reports	Misses context, images, and voice notes; high false-negative rate
Personalization	Rule-based filters	Cannot adapt to changing behavior or relationships
Explainability	Black-box ML	Users don't trust decisions they can't understand
Multimodal Understanding	Text-only classifiers	Ignores images and voice notes that carry critical signals

7. Market Opportunity

WhatsApp reports over 2 billion users globally, with a large concentration in India. Notification management and AI-driven scam detection are both active, growing categories without an established dominant player combining the two. *(Specific market-size figures such as total addressable market in dollars are directional product assumptions here, not sourced estimates — verify against a market research report before using in an investor deck.)*

Target Segments:

- Individual users — professionals, parents, students overwhelmed by notification volume
- Businesses — WhatsApp Business API users needing intelligent customer-message routing
- Enterprises — banks, e-commerce, healthcare requiring scam-safe communication channels

8. Competitive Landscape

Competitor	Strength	Weakness
Meta Scam Alerts	Native WhatsApp integration at 2B-user scale	Text-only, reactive, no personalization
On-device scam-detection apps (India-focused)	On-device processing	Limited to scam detection, no routing
Accessibility-overlay notification tools	Real-time overlay, multi-app support	Depends on accessibility-service permissions, limited AI reasoning
AI link-scanning bots	AI-powered link scanning	Bot-based, not integrated into native routing
Google Priority Inbox	Email-focused, ML-based	Email only; no multimodal or attention-budget modeling

9. Why Current Notification Systems Fail

- No attention budget — notifications are treated as unlimited rather than a limited daily allowance
- Context blindness — a payment reminder from a bank is treated the same as one from an unknown sender
- Relationship ignorance — family groups are muted uniformly, risking missed urgent messages
- Scam vulnerability — keyword filters miss image-based fraud, voice-note scams, and contextual phishing
- Black-box decisions — users don't understand why a message was notified or muted
- Static personalization — rules don't adapt to changing behavior or cognitive state

10. Product Objectives

Objective	Metric	Target
Reduce notification fatigue	Daily notifications per user	≤5 high-priority, ≤15 total
Improve scam detection	False-negative rate	<1% on multimodal scams
Increase user trust	Explainability satisfaction (survey)	≥90% understand decisions
Personalize routing	Action relevance (click-through)	≥85% acted-on notifications
Minimize false mutes	Urgent messages missed	<0.1%

11. Success Metrics

North Star: Attention Efficiency Score — ratio of high-value actions taken to total notifications received.

Secondary metrics:

- Notification budget utilization — % of daily high-priority slots used meaningfully
- Scam interception rate — % of scams blocked before the user sees them
- Explanation clarity score — user-rated understanding of AI decisions
- Relationship accuracy — % of messages correctly routed by relationship context
- Adaptive learning velocity — time to personalize to new user behavior

12. User Personas

Persona	Context & Needs
Priya, 32 — Overwhelmed Professional (Bangalore)	Buried in daily messages; misses urgent client updates amid shopping offers; wants deep-work focus with on-call
Rajesh, 45 — Family Coordinator (Hoskote)	Muted family group risks missing a child's school emergency; annoyed by payment reminders from unknown
Ananya, 21 — Student (Mumbai)	Distracted by group chats; misses payment deadlines; receives job scams; wants to balance social life with
Vikram, 38 — Small Business Owner (Delhi)	Customer messages mixed with spam on WhatsApp Business; misses urgent orders; receives fake payment

13. User Journey

Onboarding:

- Install app and grant WhatsApp notification access
- Set a daily high-priority notification budget
- Tag contacts as Family, Work, Friends, or Unknown
- Allow a short calibration period for the system to learn baseline behavior

Daily flow:

- Message arrives (text, image, or voice note)
- Multimodal engine scores attention cost, trust, and urgency
- Decision: notify now, digest later, or mute
- User sees a plain-language explanation for the decision
- User's action or dismissal feeds back into the model

14. Pain Points & 15. User Stories

User	Pain Point	Solution
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Priya	Misses urgent client messages in noise	Relationship intelligence + attention budget
Rajesh	Scam payment reminders from unknown senders	Business trust engine + scam detection
Ananya	Distracted by group chats, misses deadlines	Context engine + adaptive learning
Vikram	Fake payment links from customers	Multimodal scam detection + explainable AI

Representative user stories:

- As Rajesh, I want an urgent school message to reach me even in a muted family group, so I don't miss an emergency.
- As Priya, I want reminders from a verified bank to notify immediately while reminders from unknown senders are muted, so I avoid scams.
- As Ananya, I want a shopping offer digested rather than pushed if I've dismissed similar offers before.
- As Vikram, I want voice notes from unknown senders analyzed for scam patterns before they reach me.
- As Rajesh, I want images containing fake bill-payment links flagged before I can click them.

16. Product Differentiators & 17. Core Innovation

Differentiator	Description
Attention Budget Architecture	A daily cap on high-priority notifications, forcing prioritization
Digital Cognitive Twin	A per-user model of attention patterns and cognitive load
Relationship Graph Intelligence	Dynamic trust/urgency scoring by contact, not just content
Multimodal Scam Detection	Text, image, and voice-note fraud analysis together
Explainable Decision Timeline	Every decision carries a transparent, evidence-backed reason
Adaptive Notification DNA	The system evolves from every dismiss, act, and silence

Attention Budget Architecture: a per-user daily budget of high-priority slots resets daily; each message is scored on expected utility versus attention cost, and if the budget is full, a new candidate must displace an already-notified message — which should rarely be optimal.

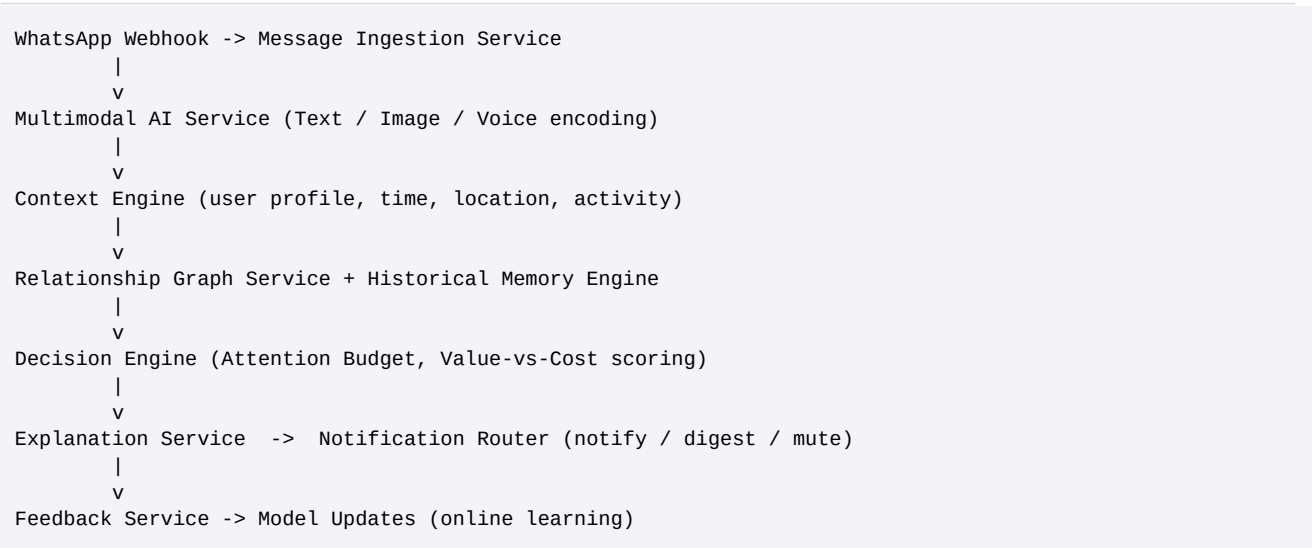
Digital Cognitive Twin: estimates current cognitive load (focused, distracted, stressed) from time, location, and activity signals, and predicts when the user is most receptive to notifications, with thresholds adapted per-user from past dismiss-vs-act behavior.

Relationship Graph Intelligence: contacts, groups, and businesses form nodes; trust, urgency, and frequency form edges; a graph-attention mechanism propagates these signals so, for example, a low-trust contact inside a high-trust group is scored differently than the same message from that contact alone.

18. Unique Selling Proposition

“Protect your attention like your bank account. AttentionOS budgets, invests, and optimizes every notification so you never miss what matters.”

19–21. Product, Technical & AI Architecture



Stack (as proposed for a production build): React Native (mobile), Next.js (dashboard), Node.js + FastAPI (backend), PyTorch/Hugging Face Transformers (AI pipeline), PostgreSQL + Redis + Neo4j (data), S3 + Pinecone

(media & embeddings), AWS/GCP with Kubernetes (infrastructure). See the Implementation Status appendix for what the 24-hour hackathon build actually runs on.

22. AI Pipeline (Proposed)

- Ingestion — extract message_id, sender_id, group_id, timestamp, content
- Multimodal encoding — text via a language encoder, images via a vision encoder, voice via speech-to-text plus an audio encoder
- Context encoding — user profile, relationship graph, historical memory combined into a context vector
- Fusion & scoring — combine embeddings, propagate relationship signals, apply the attention-budget constraint, compute value-vs-attention-cost
- Decision head — outputs action (notify/digest/mute), confidence, reason, evidence references
- Explanation generation — a language model produces a concise, human-readable reason
- Feedback loop — user's act/dismiss updates the models

23–33. Engine Details (Condensed)

Context Engine: combines user profile, time, location, and activity into a context vector; estimates cognitive load and predicted receptivity; weights messages contextually (e.g. work messages weighted higher during work hours).

User Intelligence Layer: a per-user model of attention patterns; tracks daily budget usage; learns thresholds from dismiss-vs-act history; schedules digests for receptive moments.

Relationship Intelligence Layer: a graph of contacts/groups/businesses with trust, urgency, and frequency as edge weights, updated dynamically from interactions; trust tiers span trusted, unknown, and suspicious.

Business Trust Engine: verifies business senders, scores trust from history and user reports, and flags unusual requests (money, gift cards, PINs) or rushing language.

Historical Memory Engine: a vector store of past message embeddings plus an action log, used to retrieve similar prior messages and the user's past response as evidence for new decisions.

Scam Detection Engine: multimodal analysis across text (phishing language, urgency, money/PIN requests), images (fake bills, malicious QR codes/links), and voice notes (transcription plus phishing-pattern detection); outputs a 0–1 scam score.

Explainable AI Engine: generates a local explanation per decision plus aggregate, system-level explanations of overall behavior, with evidence references attached to each.

Decision Intelligence Engine: enforces the attention-budget constraint and computes value-vs-attention-cost to produce the final action, confidence, and reason.

Adaptive Learning Engine: updates thresholds and relationship weights online from user feedback — e.g. raising the threshold for a message category after repeated dismissals.

34. Notification Decision Matrix (Illustrative)

Message Type	Trust	Urgency	Budget	Action
Bank payment reminder	High	High	Available	Notify
Unknown payment reminder	Low	Medium	Available	Mute (scam risk)

Family group message	High	Low	Full	Digest
Family emergency keyword	High	High	Full	Notify (displaces)
Shopping offer (dismissed before)	Medium	Low	Available	Digest
Verified business order	High	High	Available	Notify

35–36. Functional & Non-Functional Requirements

- FR — receive messages via webhook; encode text/image/voice; score attention cost, trust, urgency
- FR — enforce the daily high-priority notification cap; route to notify/digest/mute
- FR — generate human-readable explanations; store evidence references
- FR — learn from user feedback; detect scams multimodally; verify business senders
- NFR — target latency under ~500ms per message; high availability; horizontal scalability
- NFR — on-device processing where feasible for privacy; encryption for stored data
- NFR — every decision carries a reason; personalization should converge within about a week of use

37–39. APIs, Database & Storage (Proposed)

Proposed integrations: WhatsApp Business API (ingestion, business verification), a location API for context, a calendar API for activity context, and vector/graph stores (e.g. Pinecone, Neo4j) for embeddings and relationship data.

Proposed schema: Users (budget, cognitive state, preferences), Messages (content, media references, action, confidence, reason, evidence IDs), Relationships (trust, urgency, frequency per contact), Actions (per-message user response log).

40. AI Models (Proposed)

Component	Approach
Text encoder	Transformer-based language encoder — text embeddings
Image encoder	Vision-transformer/contrastive model — visual embeddings
Voice encoder	Speech-to-text plus audio-feature model — transcription and audio embeddings
Multimodal fusion	Transformer fusion layer — combines text/image/voice
Graph attention	Graph attention network — relationship scoring
Decision head	Lightweight classifier — action, confidence, reason
Explanation generator	An instruction-tuned language model — human-readable reasons
Scam classifier	Fine-tuned text classifier — phishing detection

41–44. Prompting, Retrieval & Scoring (Proposed)

Explanation generation is proposed as a templated prompt that supplies the message, sender, trust/urgency scores, similar past messages, and action history, and asks for a concise (max two-sentence) reason referencing trust, urgency, and past behavior.

Evidence retrieval is proposed via vector similarity search over past messages, filtered toward cases where the user actually acted rather than dismissed.

Confidence is proposed as a sigmoid function of $(\text{value score} \times \text{trust score} \times \text{urgency score} - \text{attention cost})$, calibrated so confidence above ~ 0.8 notifies, $0.5\text{--}0.8$ digests, and below 0.5 mutes.

45–52. UI/UX, Demo Flow & Architecture Diagrams

Concept: a notification card surfaces the message, an action button, and its explanation together — e.g. “Notified: payment reminder from a trusted, verified sender.” A dashboard shows budget usage, scam interception rate, and the relationship graph.

Proposed demo flow:

- Onboarding — set a daily notification budget, tag a few contacts by relationship
- Message simulation — a trusted payment reminder (notify), a dismissed-before shopping offer (digest), an unknown-sender voice note (mute, flagged)
- Dashboard walkthrough — budget usage, scam interception rate, relationship graph
- Feedback loop — dismissing a digest message and showing the system adapting

53–55. Security, Privacy & Ethical AI Considerations

- Encrypt data in transit and at rest; process sensitive content on-device where feasible
- Minimal data collection; user-controlled deletion and export; clear explanations for every decision
- Regular fairness audits on scam detection and relationship scoring; diverse training data
- No optimization for engagement — the budget constraint is a hard cap, not a suggestion

56–59. Scalability, Performance & Failure Handling

Proposed scaling approach: independently-scalable microservices, read replicas and sharding for the database layer, model distillation for edge/on-device scam detection, and batch inference during peak load. Failure handling proposes graceful degradation to rule-based routing if the AI service is unavailable, with local queuing during connectivity loss.

60. Future Roadmap (Proposed Phasing)

- Phase 1 — MVP: core pipeline, attention budget, relationship graph, mobile app
- Phase 2 — Scaling: multimodal scam detection, business trust engine, feedback loop
- Phase 3 — Expansion: cross-platform support, enterprise features, third-party API
- Phase 4 — Long-term vision: a broader cross-app attention model, deeper platform partnerships

61–63. Business Model & Commercialization (Proposed)

Proposed revenue streams: a freemium consumer tier (basic routing free, an unlimited/advanced-scam-detection paid tier), a per-message or enterprise-volume WhatsApp Business API pricing model, and API licensing to other messaging platforms. *Specific price points, CAC/LTV figures, and TAM numbers in the original draft are illustrative placeholders and should be validated against real acquisition data before use in an investor conversation.*

Proposed go-to-market starts in high-WhatsApp-penetration markets before expanding globally.

64–70. Pitch Summary

Elevator pitch: AttentionOS treats attention as a budget rather than an inbox — a small daily allowance of high-priority notifications, with everything else routed to digest or mute based on relationship, trust, and history, and every decision explained in plain language.

Technical pitch: a multimodal pipeline (text, image, voice) combined with relationship-graph scoring and an attention-budget constraint, producing an explainable notify/digest/mute decision with evidence — proposed to run on a modern ML stack in production, with a scoped rule-engine-plus-LLM prototype for the hackathon (see appendix).

Business pitch: a drop-in intelligence layer for existing messaging infrastructure rather than a new app to adopt, addressing both personalization (fatigue reduction) and trust & safety (scam detection) — two active, underserved demands in the same product.

Closing vision: a future where attention is protected the way money is — budgeted, invested, and optimized — shifting the notification layer of messaging apps from “notify on arrival” to “notify on judgment.”

Appendix — Hackathon Implementation Status

Added for judging clarity: this appendix distinguishes what the 24-hour prototype actually runs, versus the production vision described above.

Capability	Status
Rule-based routing (notify / digest / mute)	Implemented
Scam / phishing floor across all senders	Implemented
@-mention mute bypass	Implemented
Multimodal voice + image preprocessing (Gemini)	Implemented
Urgency keyword elevation	Implemented
Explainable decision trace	Partial
Relationship scoring (static lookup, not a learned graph)	Partial
Attention Budget (decaying daily threshold)	Roadmap
Digital Cognitive Twin (evolving per-user profile)	Roadmap
Neo4j relationship graph / Pinecone semantic memory	Roadmap
Production stack (Kubernetes, PostgreSQL, PyTorch models)	Roadmap — prototype runs on Gemini + a rule engine